



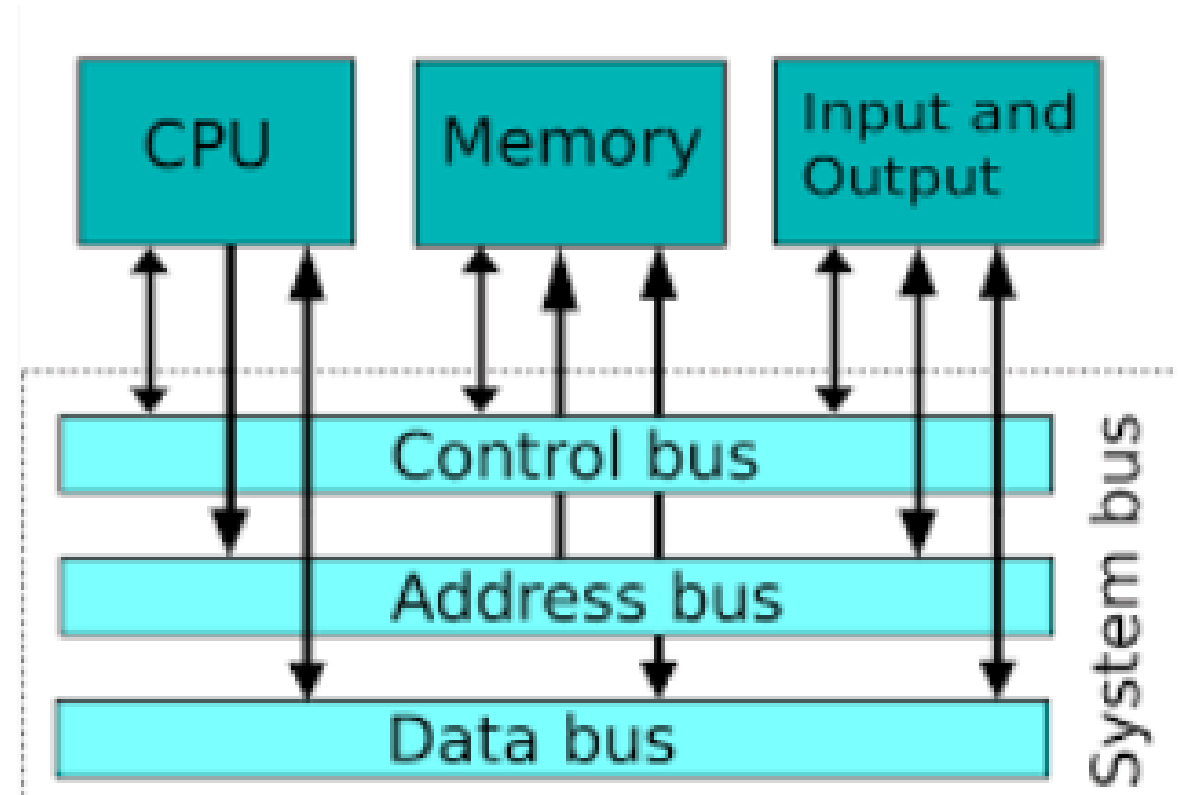
CSC 105 Computer Architecture and Organization

1st Semester, SY 2026-2027

System Bus and Interconnection Network

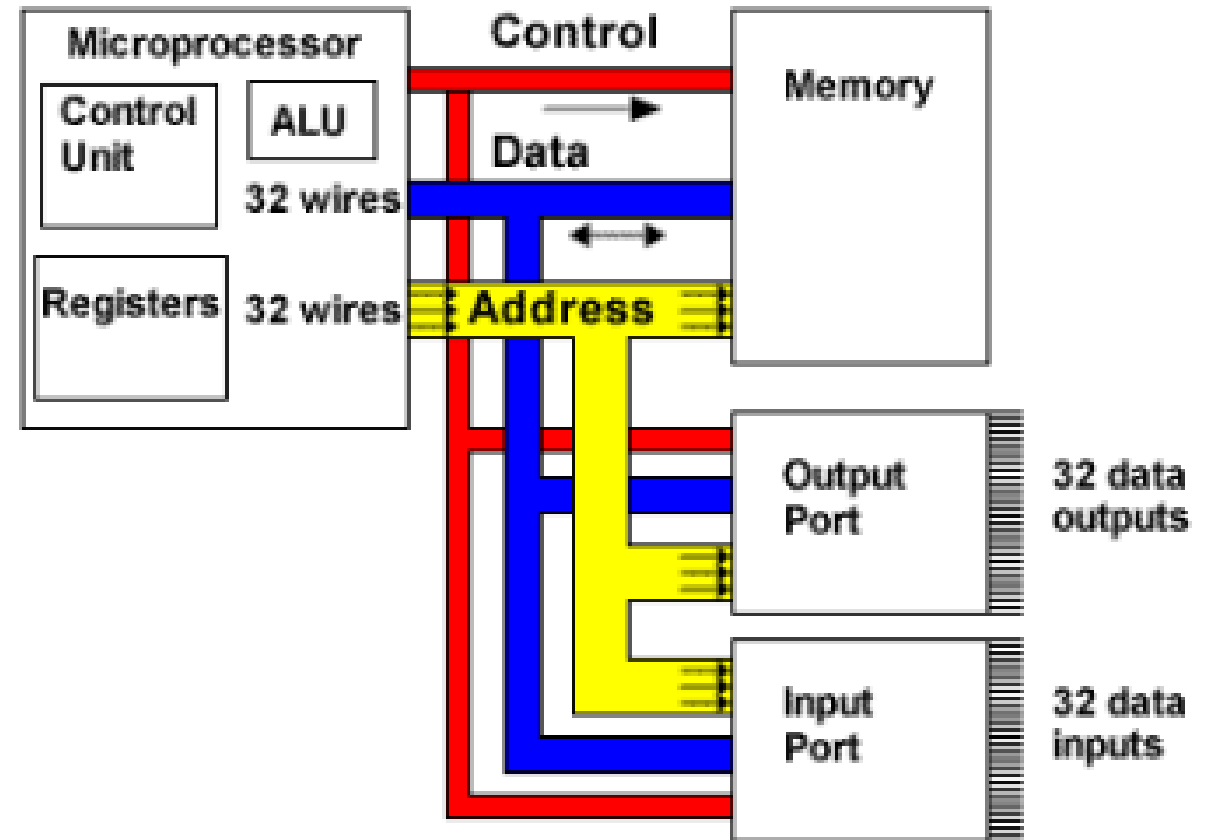
What is a Bus?

- A communication pathway connecting two or more devices
- more devices share the same bus
- Often grouped
 - A number of channels in one bus
 - e.g. 32 bit data bus is 32 separate single bit channels
- system bus is divided into three main categories
 - Control bus, Address bus and Data Bus

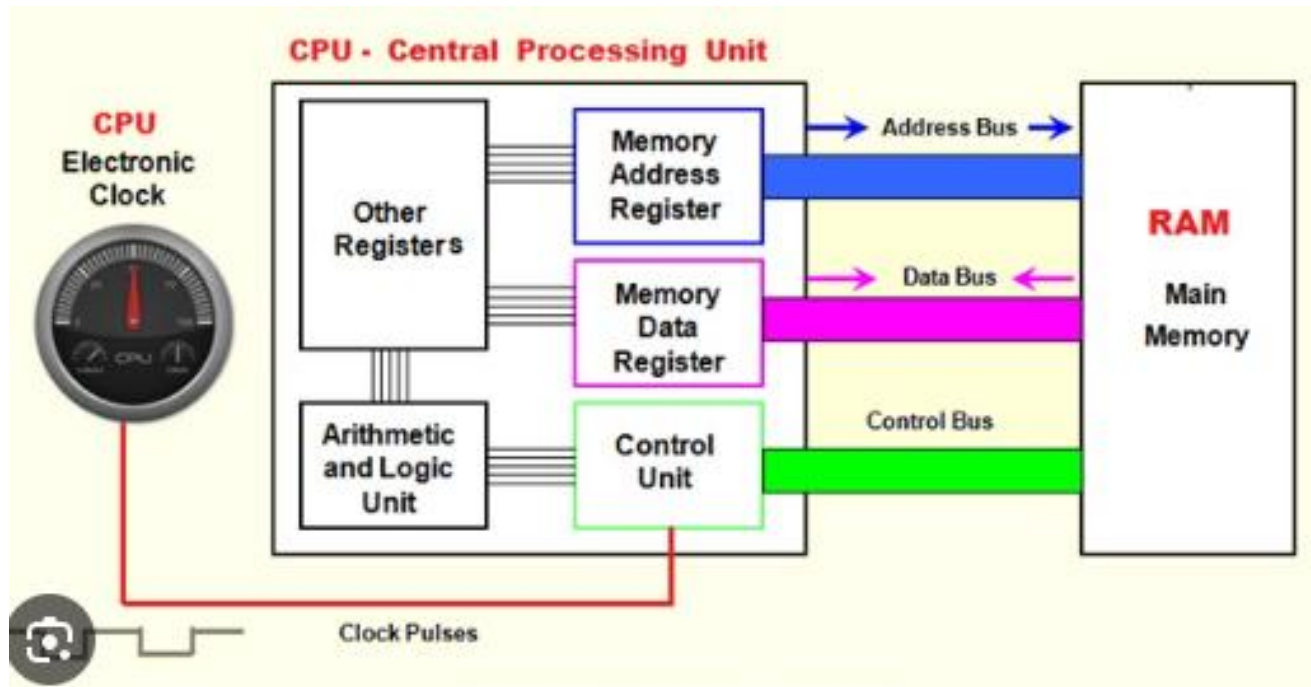


Data Bus

- Carries data
 - Remember that there is no difference between “data” and “instruction” at this level
- Width is a key determinant of performance
 - 8, 16, 32, 64 bit

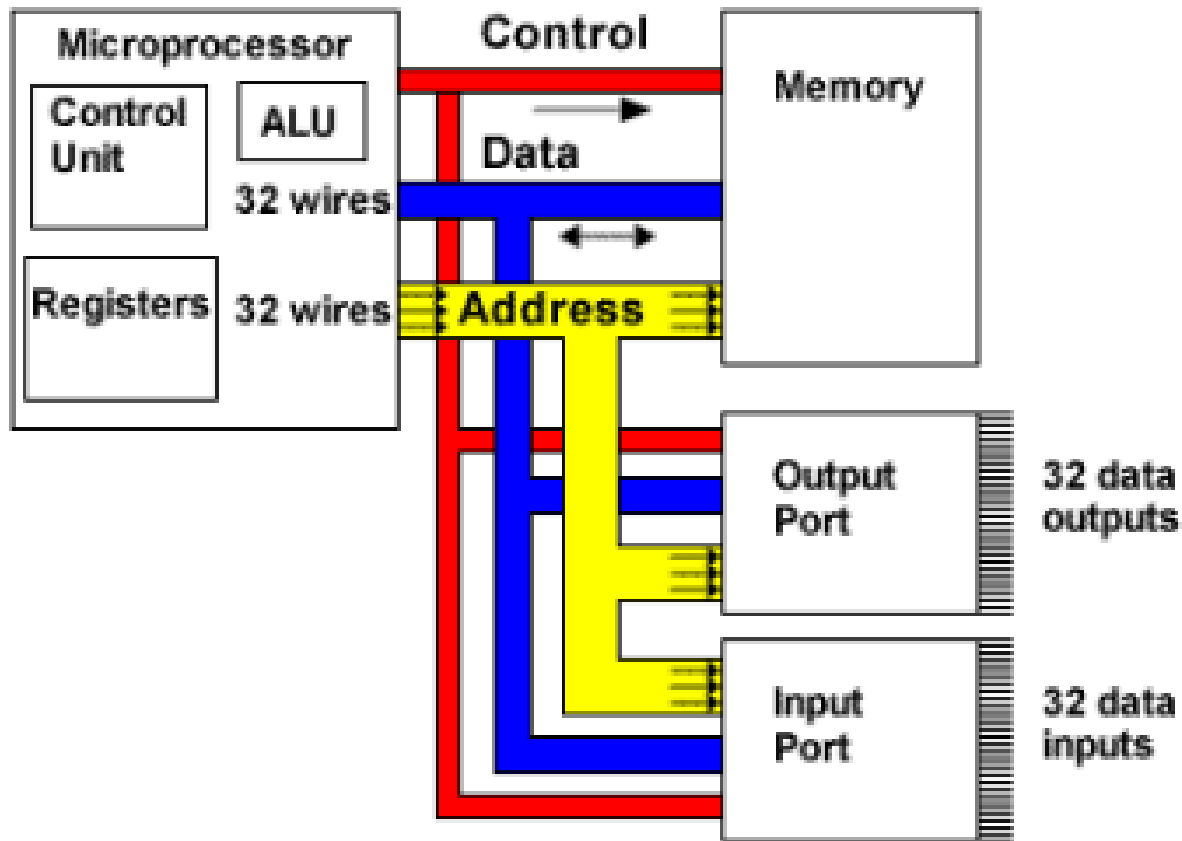


Data Bus Operation



How it works?

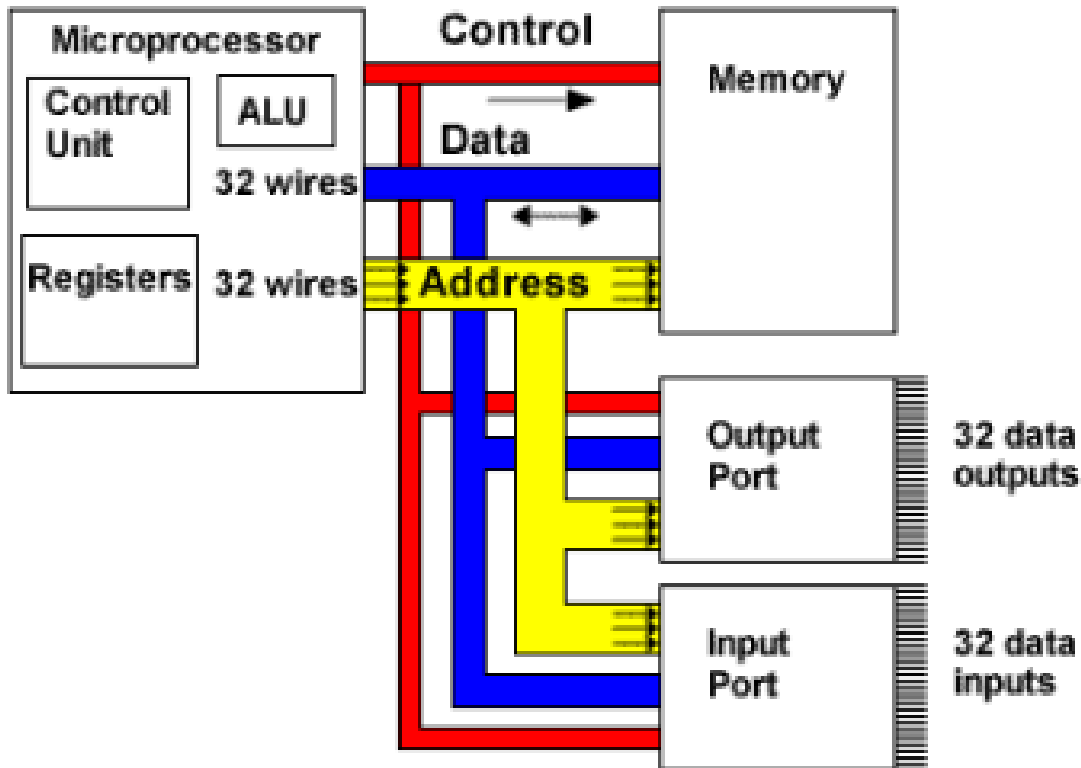
- Reading Data: When the CPU needs to retrieve data, the data bus carries that information from the memory (like RAM) or an input device towards the CPU.
- Writing Data: Conversely, when the CPU sends data to be stored or to an output device, the data bus carries this information away from the CPU.
- Why bidirectional?
- The data bus must support both reading and writing operations. The CPU needs to receive data from memory, but it also needs to send data to memory or output devices



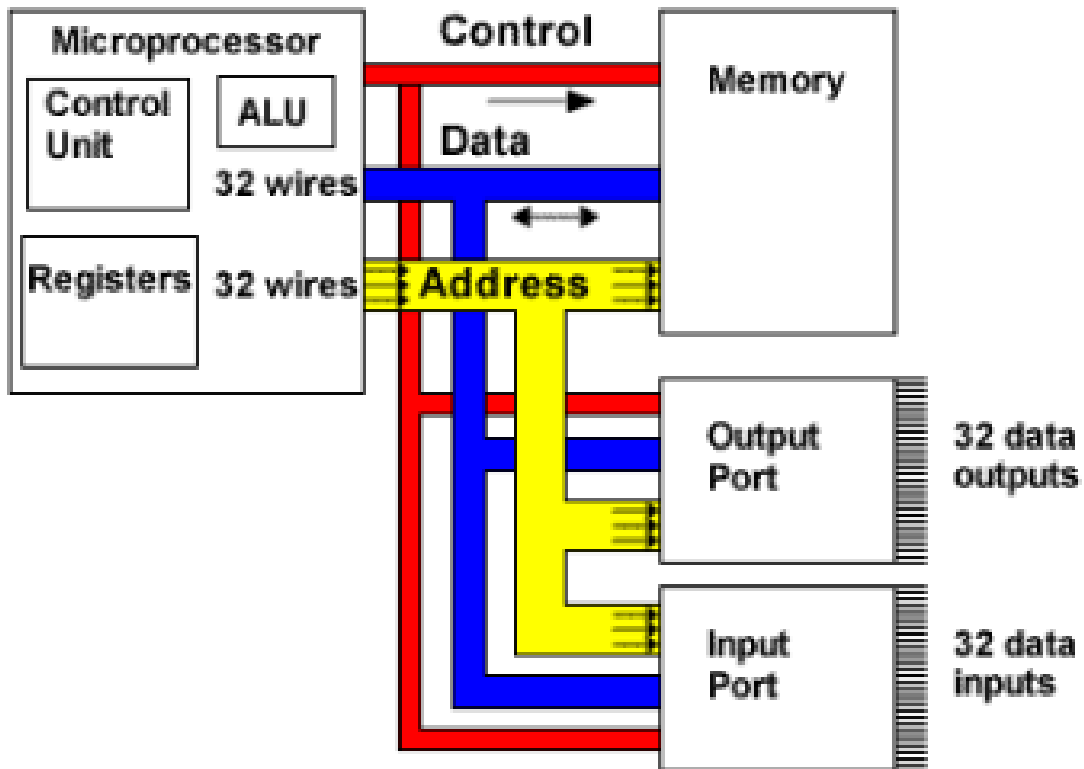
Address Bus

- Identify the source or destination of data
- e.g. CPU needs to read an instruction (data) from a given location in memory
- Bus width determines maximum memory capacity of system
 - e.g. 8080 has 16 bit address bus giving 64k address space

Address Bus Operation

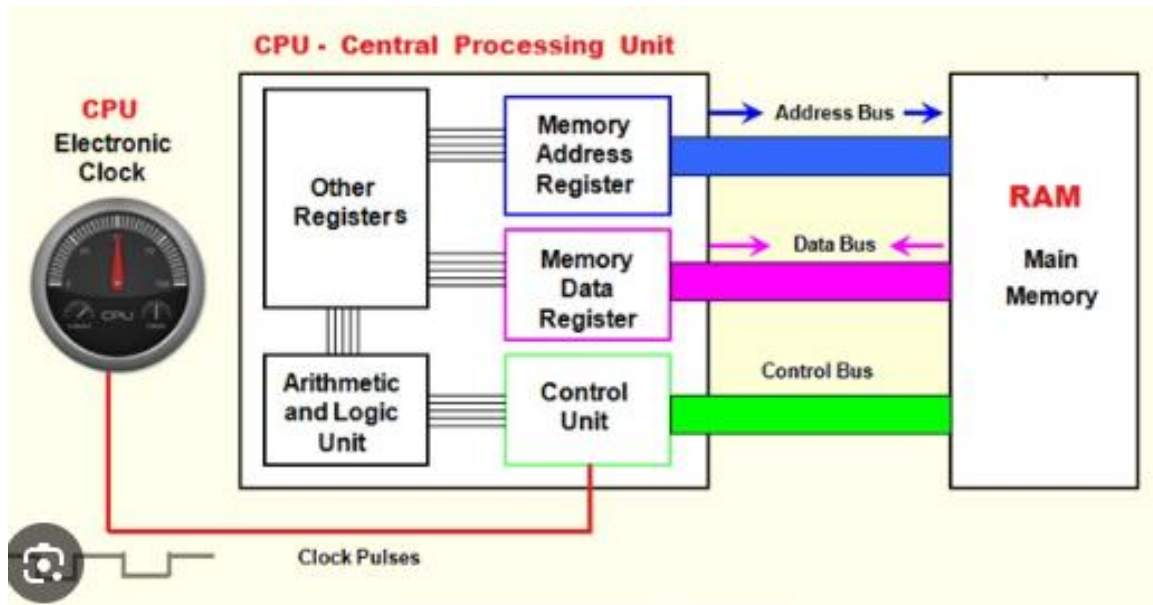


- Key Functions of an Address Bus
- Memory Location Specification: When the CPU needs to access data, it places the precise memory address on the address bus.
- Unidirectional Communication: The address bus is typically unidirectional, meaning data flows only from the CPU to other components, not the other way around.
- Memory Capacity: The number of bits on the address bus directly correlates to the maximum amount of memory that can be addressed.
 - A 1-bit address bus can access $2^1 = 2$ locations.
 - A 32-bit address bus can access 2^{32} locations.



How it Works with Other Buses?

- **Data Bus:** While the address bus carries the location, the data bus carries the actual data to and from that memory location.
- **Control Bus:** The control bus sends commands (like read or write requests) that coordinate with the address on the address bus to complete an operation.

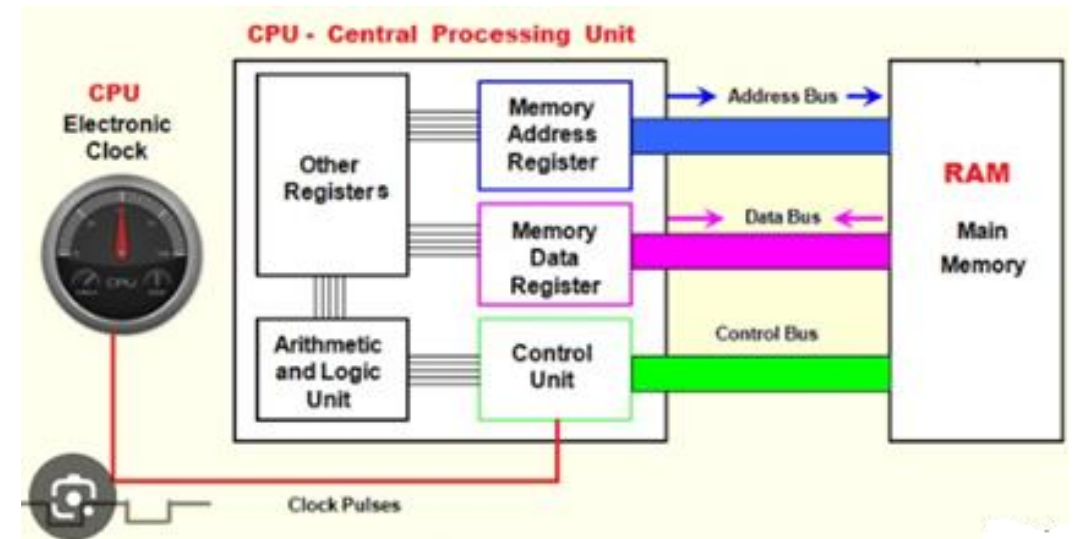
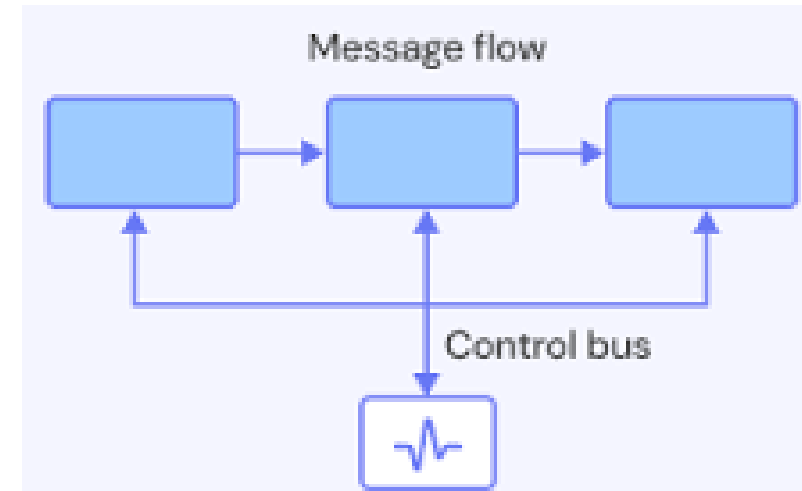


Control Bus

- A control bus is a set of bidirectional communication pathways within a computer's system bus that transmits control and timing signals between the CPU and other hardware components like memory and I/O devices
- Control and timing information
 - Memory read/write signal
 - Interrupt request
 - Clock signals

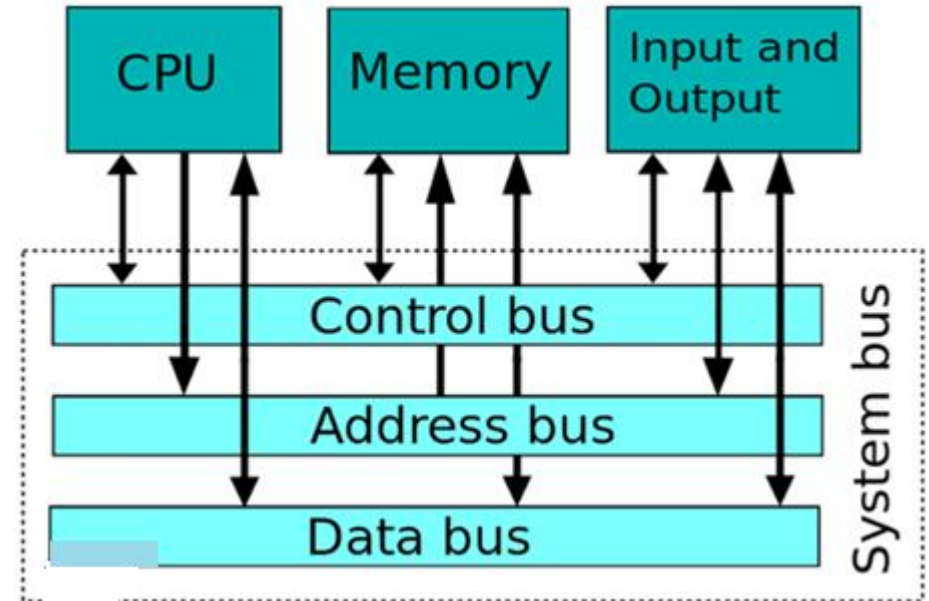
Control Bus Operation

- Directs Operations: The control bus is responsible for coordinating and synchronizing the activities of various hardware components.
- Carries Control Signals: It transmits signals like:
 - Read/Write Requests: To tell memory or I/O devices whether to read data or write data.
 - Interrupts: Signals from devices to the CPU to indicate they need attention or have a request.
 - Clock Signals: To synchronize operations across different components.
 - Reset: A signal to restart or reboot the CPU.



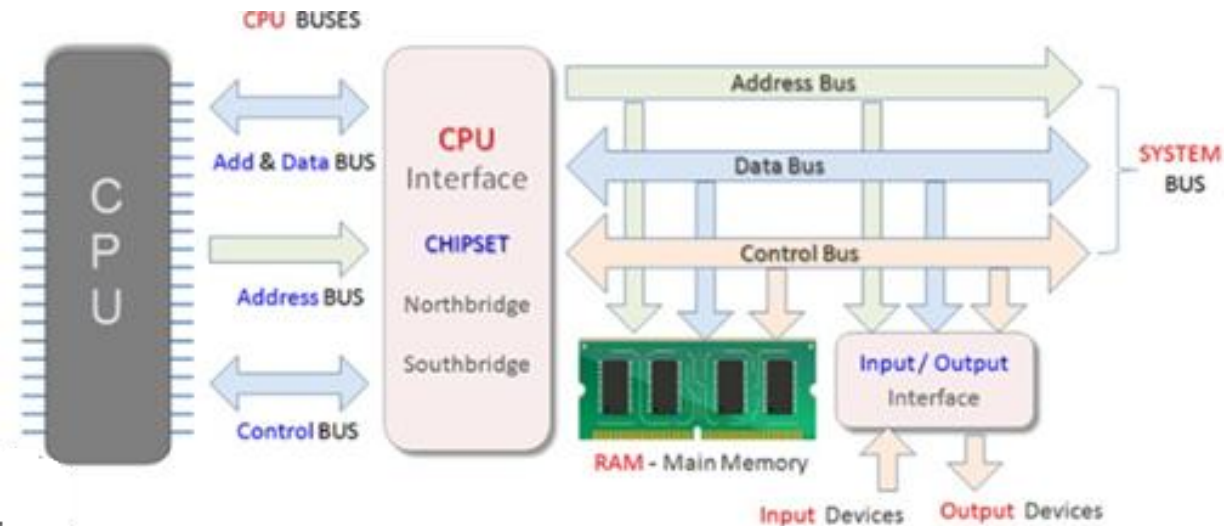
How does control bus works? (cont'n)

- CPU Initiates an Operation: When the CPU wants to fetch an instruction or save data, it sends a command through the control bus.
- Device Responds: The target device (like memory) receives the command and initiates the action, such as reading or writing data.
- Status Feedback: After the operation is complete, the device can send a status signal back to the CPU via the control bus

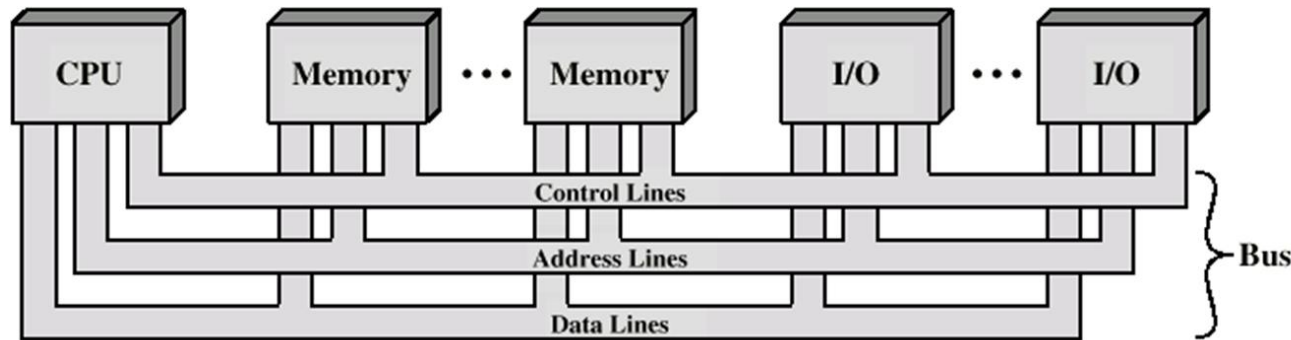


Control bus relationship with other buses

- The control bus works in conjunction with two other primary buses to form the system bus:
 - Data Bus: A bidirectional pathway that transfers actual data between components.
 - Address Bus: A unidirectional pathway that specifies the physical location in memory or an I/O device to be accessed



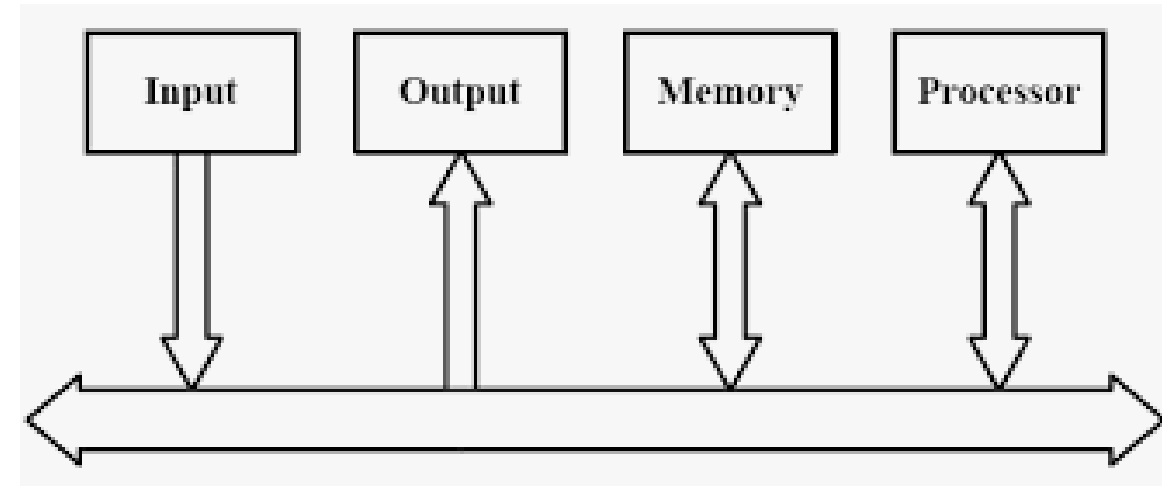
Bus Interconnection Scheme



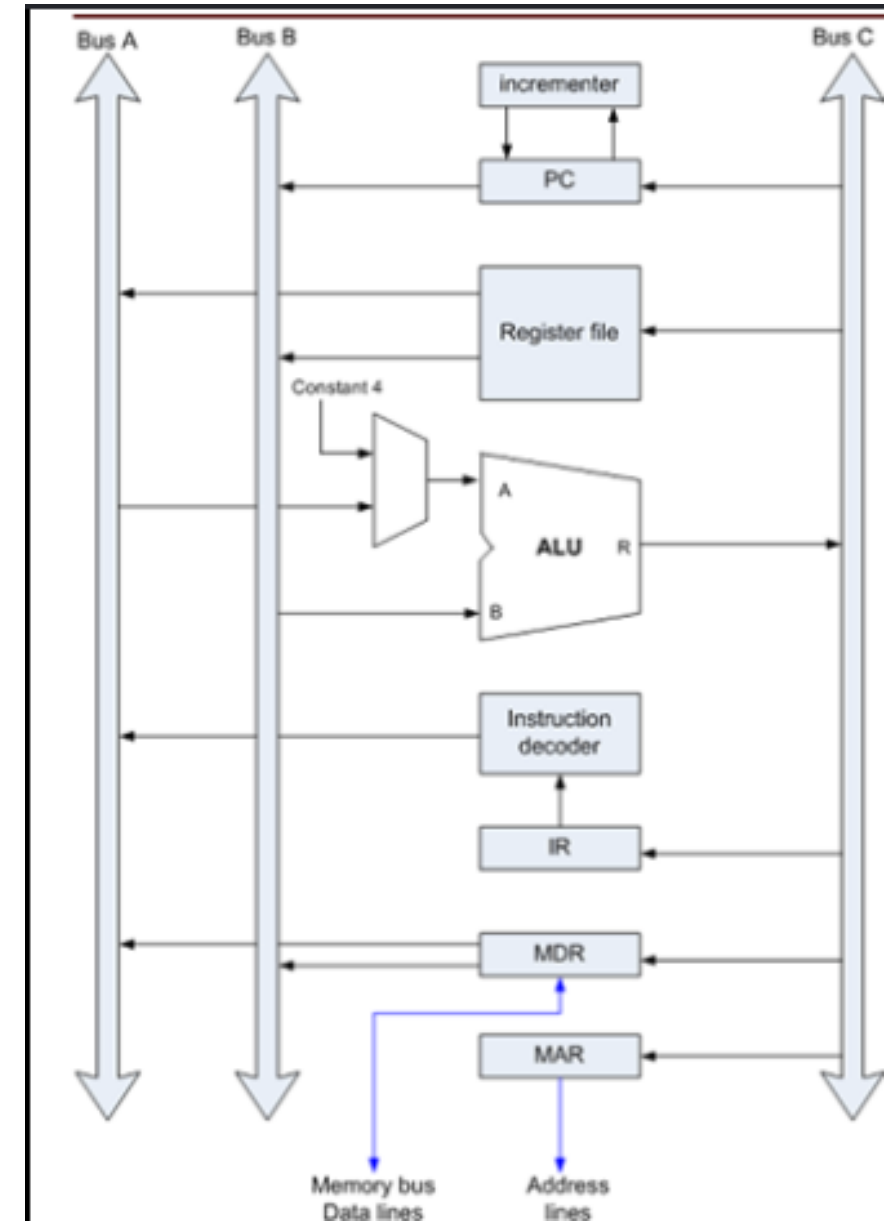
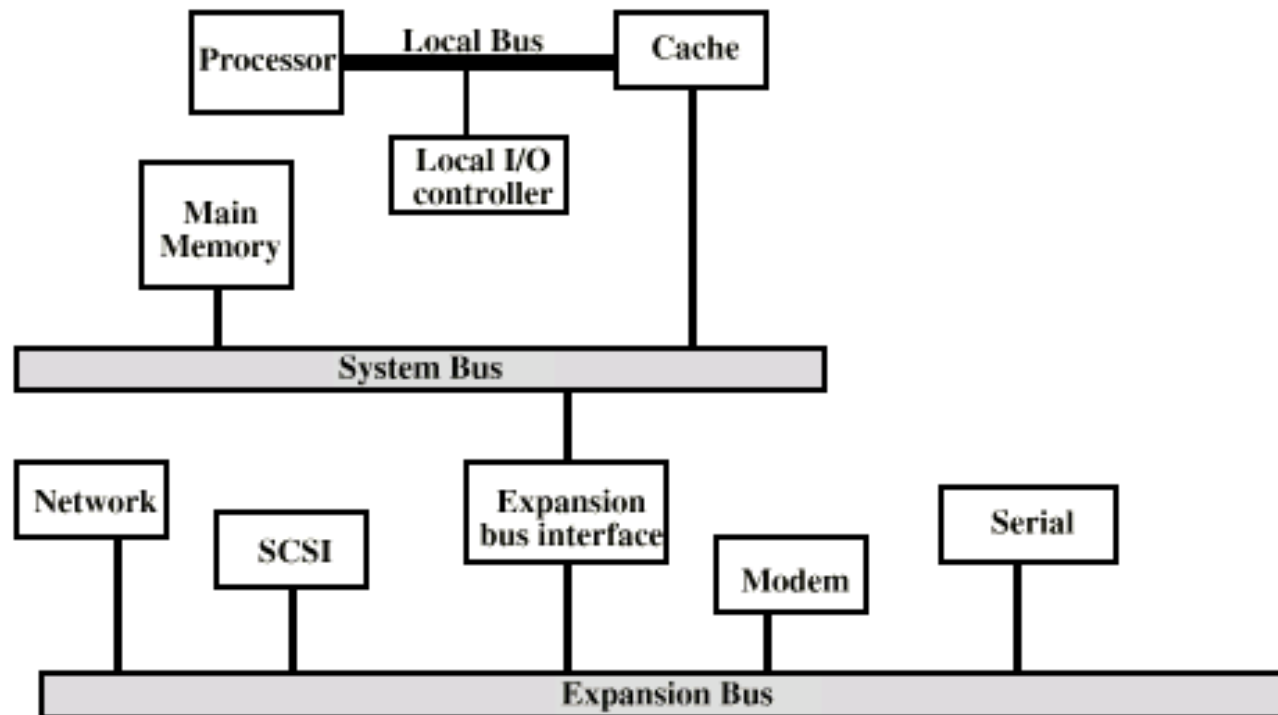
- Every device is attached to the bus:
 - its use needs to be coordinated
- How does it work?
 - Device Connection: Multiple devices, such as the CPU, memory controllers, and I/O units, connect to the bus.
 - Request for Access: When a device needs to send or receive data, it initiates a request to access the bus.
 - Bus Arbitration: A mechanism called bus arbitration determines which device gets to use the bus and when, ensuring that only one device transmits at a time.
 - Data Transfer: The winning device then transmits the address and/or data on the address and data lines, respectively, along with appropriate control signals, allowing the target device to receive the information.

Single Bus Problem

- Lots of devices on one bus leads to:
 - Propagation delays
 - Long data paths mean that coordination of bus use can adversely affect performance
 - If aggregate data transfer approaches bus capacity
- Most systems use multiple buses to overcome these problems

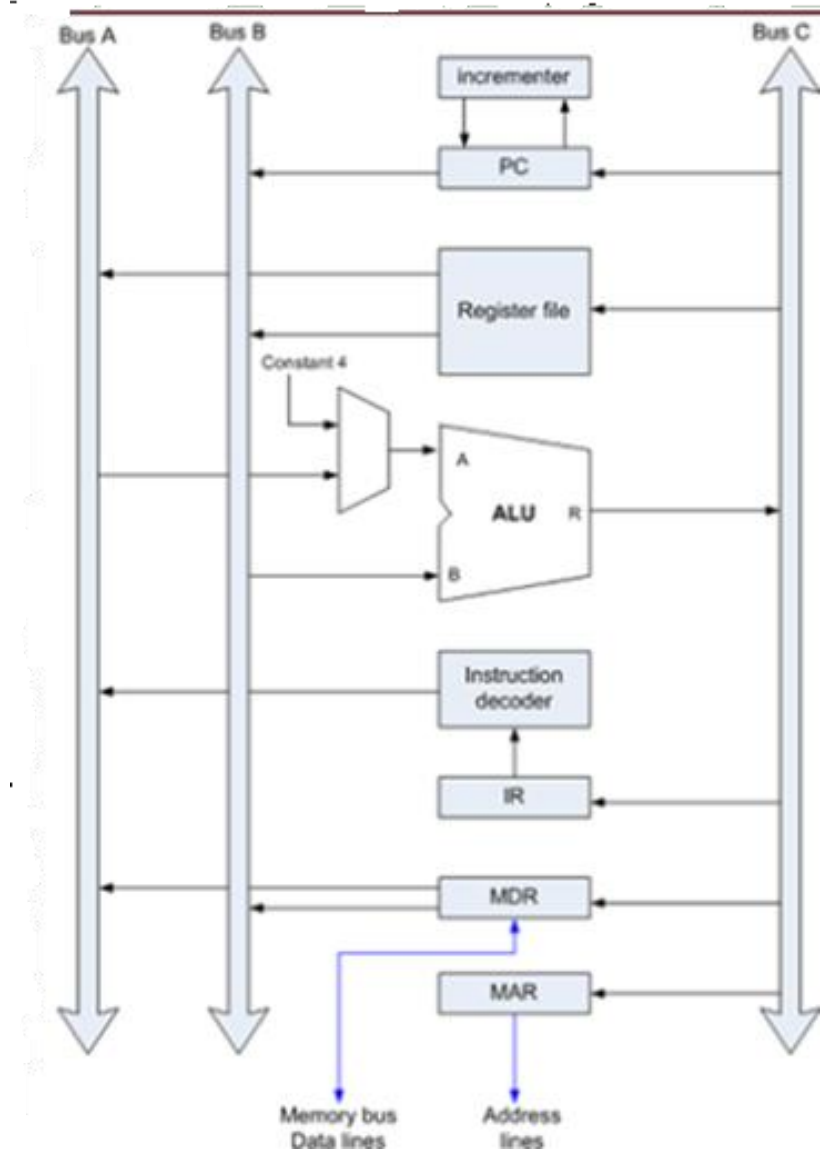


Multiple-Bus Architecture



Multiple-bus Architecture Operation

- **Parallel Operations:** Multiple independent buses allow for simultaneous data transfers and operations between different components, like fetching two operands for the ALU and sending the result back to the register file in a single clock cycle.
- **Dedicated Paths:** In system-level architectures, different buses serve specific roles. A local bus might connect the CPU to the cache, while a system bus handles main memory and a high-speed bus manages I/O devices like graphics cards.
- **Reduced Contention:** By segmenting traffic onto different buses, the likelihood of multiple devices trying to access the same bus at the same time is minimized, leading to smoother and more efficient data flow.



Benefits of Multiple-Bus Architecture

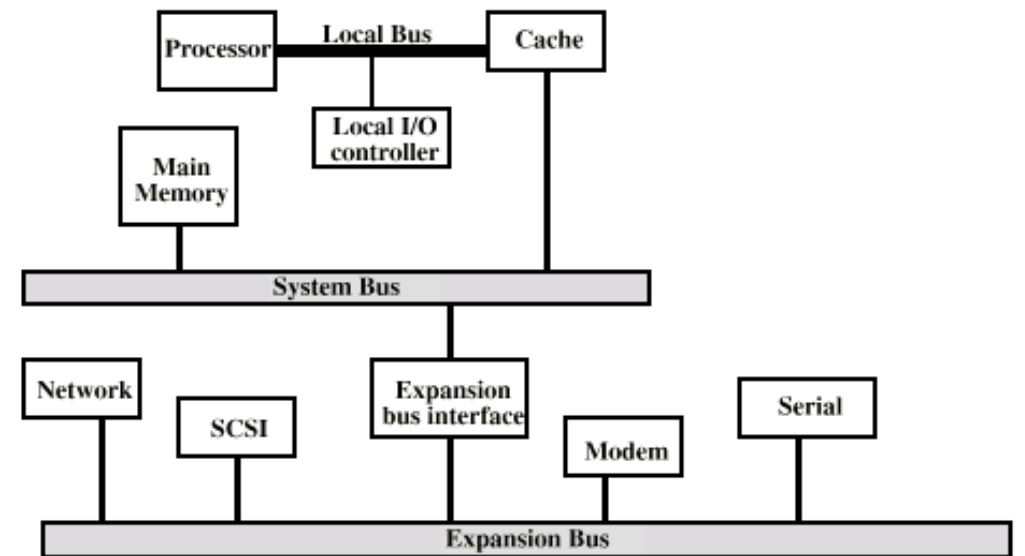
- **Increased Speed:** Simultaneous data transfers and operations lead to faster program execution.
- **Higher Throughput:** The processor can handle more operations or jobs in a given amount of time by processing them concurrently.
- **Enhanced Efficiency:** Specialized buses ensure that each component gets the resources it needs without having to wait for others on a shared path.
- **Improved Resource Utilization:** Different components, such as the ALU and registers, can be better utilized when they have dedicated paths for communication

Drawbacks

- Increased Complexity: Designing a system with multiple interconnected buses is more intricate than a single-bus design.
- Higher Cost: More buses, wiring, and components increase the overall cost of the system.
- Challenging Maintenance: The increased number of parts and connection points can make troubleshooting and maintenance more difficult

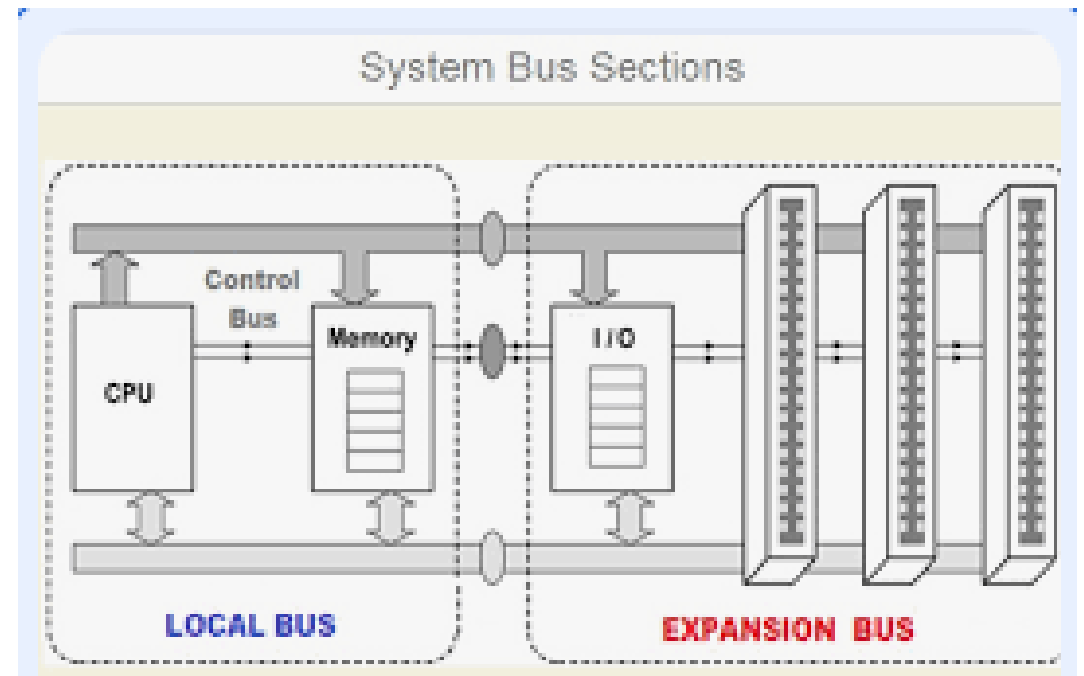
Types of Bus

- Local Bus - A high-speed, high-bandwidth bus that connects internal components such as the CPU, memory, and graphics cards.
- System Bus - The collective term for the various pathways that allow all the components of a computer to communicate. It can specifically refer to the main, high-speed bus that links the CPU, memory, and other core components.
- Expansion Bus - An external bus designed to connect peripherals and add-on devices to the motherboard



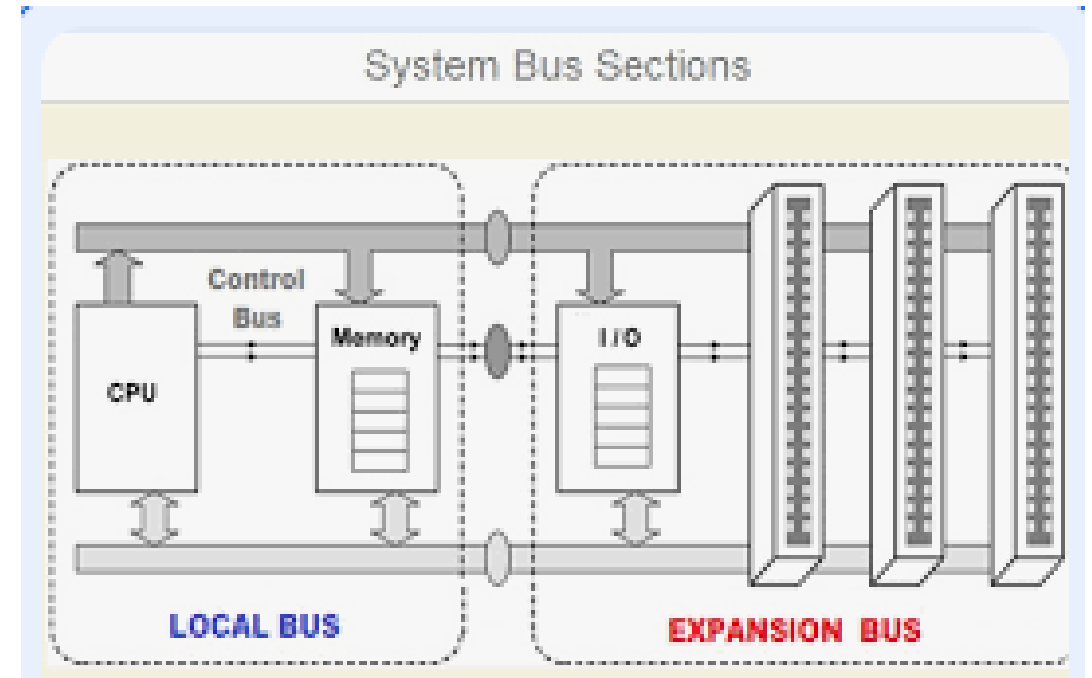
LOCAL BUS

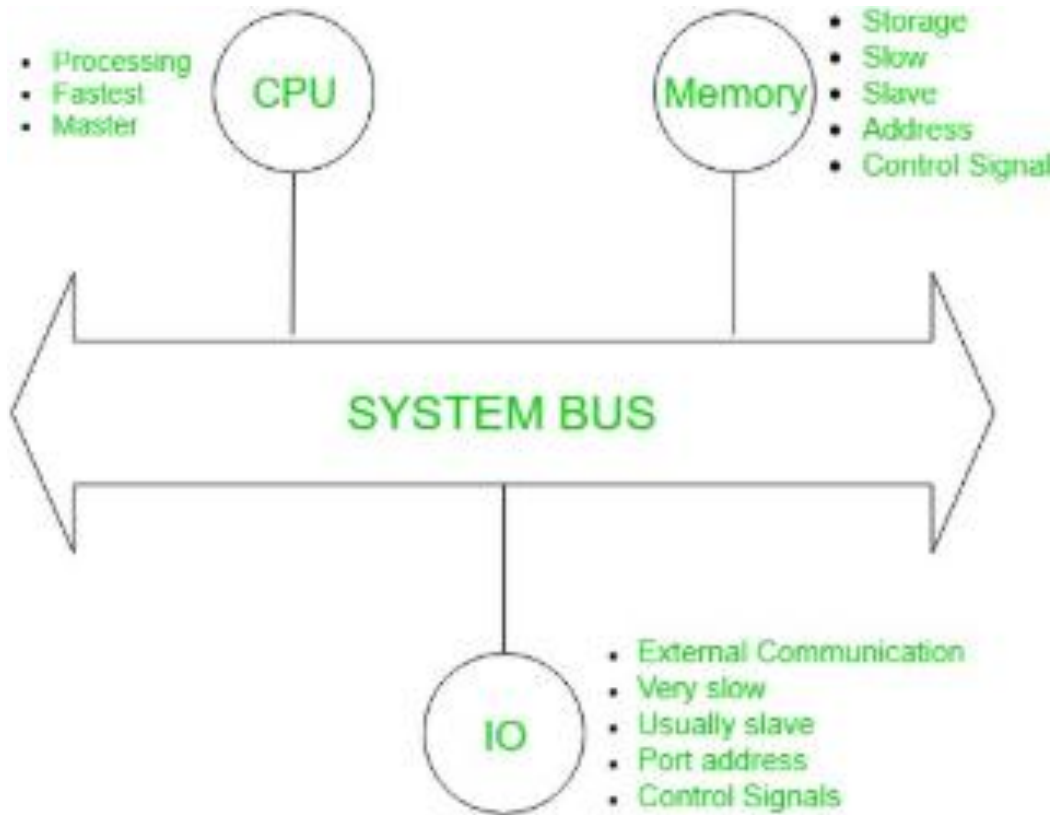
- A local bus (or internal bus) is a high-speed bus connecting the core internal components of a computer, like the CPU and memory, to the motherboard.
- Purpose: To allow fast, direct communication between critical components within the computer.
- Examples: A dedicated bus for the graphics card or the bus connecting the CPU to the main memory (RAM).



EXPANSION BUS

- An external bus designed to connect peripherals and add-on devices to the motherboard.
- Purpose: To provide a flexible pathway for adding new functionality or connecting external devices to the computer.
- Examples: USB, PCI (Peripheral Component Interconnect), and older buses like ISA (Industry Standard Architecture)





SYSTEM BUS

- The collective term for the various pathways that allow all the components of a computer to communicate. It can specifically refer to the main, high-speed bus that links the CPU, memory, and other core components.
- Purpose: To provide a complete communication infrastructure for the entire computer system.
- Key Characteristic: The system bus often serves as the central hub from which other buses, including local and expansion buses, branch out to connect different components.